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Chapter 4

Job Management Overview



Introduction

Job Management covers the requirements of the wide range of companies that need to capture costs and income on specified projects. Such jobs may range from a large project such as a multi-site building scheme which needs to be broken down into various jobs and tasks with stage payments, to a one-off commission where a consultant is required by a client to produce a time and material invoice.

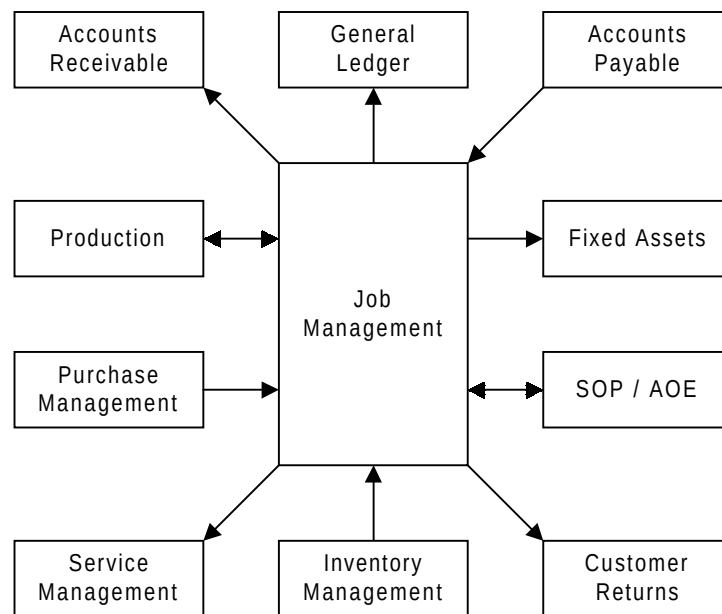
Job Management supports both multi-company and multi-currency processing, providing the necessary information and facilities in the following key areas:

- Quotations - once created, these can be converted into jobs.
- Job entry / amendment - set up fixed price, time & materials, or non-chargeable jobs.
- Flexible job hierarchy enables costs and income to be recorded at all levels.
- Costs and income can be posted to the General Ledger; assets can be created in Fixed Assets.
- Cost recording at any job level is by user-defined expense type.

Costs are captured via:

- Entry of employee timesheets.
- Entry of employee expenses.
- Direct entry of user-defined transactions.
- Entry of standing costs.
- Costs from other applications: Accounts Payable, Purchase Management, Inventory Management and Production.
 - Budgets at all job levels by period.
 - Automatic invoicing (for chargeable jobs), with the facility to review and amend prior to printing.
 - Flexible enquiry facilities .

The Environment



General Ledger

Detailed cost and income values are held in the Job Management system while a financial summary of job values is provided in the General Ledger (GL). Obviously, it is imperative that the two sets of values are balanced and can be reconciled.

A Job Management sub-ledger can be set up within the General Ledger with its own control account for work in progress control, as well as other holding accounts for costs entered within Job Management, Inventory Management and Accounts Payable. This is optional.

Each transaction (cost or income) is posted to a valid General Ledger account which must already be set up. Posting masks enable you to target a GL account without having to enter a code on every transaction.

Accounts Receivable

Job Management shares customer detail information with Accounts Receivable. Open items are created and relevant accounts are updated on printing of invoices.

Production

You can create production orders for jobs within Job Management.

Purchase Management

Purchase Management accumulates the value of outstanding purchase orders, with matching job numbers, against each job level. You can use Purchase Management to display the outstanding orders for a job.

Accounts Payable

Costs attributed to outside sources such as sub-contractor costs, bought out materials and services, can be entered into the Job Management system via Accounts Payable. This occurs when the invoice is accepted into the Accounts Payable system by a request for additional information, i.e. job / sub-job / task, plus a breakdown of the type of expense. Employee expenses can also be entered through Accounts Payable.

Inventory Management

Inventory Management is used for issuing physical stock on an ad hoc basis.

Fixed Assets

The interface to Fixed Assets enable fixed assets to be held against more than one job detail. Where projects using Job Management involve the construction or assembly of fixed assets, the transfer of the asset from Job Management to fixed assets can be completed via a transfer facility within Job Management.

Sales Order Processing / Advanced Order Entry

Job Management can interface with Inventory Management to capture the cost of inventory items. Job Management captures the necessary costs by defining a list of parts needed for a job detail. It interacts with Advanced Order Entry or Sales Order Processing to define the parts list.

Sales Order Processing progresses the list of parts, defined as an order, from entry through to despatch. This facilitates the production of acknowledgements, pick notes and 'clean' despatch notes for the parts list. Because the invoices are produced within Job Management, confirmation of despatch prevents the creation of an invoice – instead the inventory costs are posted to Job Management. Each job detail (i.e. any job / sub-job / task number) may be assigned to one sales order.

Customer Returns

Job Management allows access to the Customer Return functions of:

- adding or amending a return
- receiving a return

This enables returns to be linked to a job.

Service Management

Job Management will inform you if installation records deriving from a parts list need attaching to a contract, and it can provide access to Service Management to perform the attachment.

Two Core Principles

Job Hierarchy

To provide complete flexibility for diverse types of projects and reporting requirements, the definition of a job is divided into three levels:

- Main job - 6 characters long.
- Sub-job - 4 characters long.
- Task - 3 digits long.

The main job is mandatory, but the entries for sub-job and tasks are optional depending on the complexity of the project and the amount of detail required for control and analysis.

Roll-Up

Costs - Cost transactions are entered against a job by specifying the main job, the sub-job and task (if required), plus the expense type. Costs and income are 'rolled up' (i.e. transferred) to the next level. The system will hold the 'this level' costs and income, as well as the rolled-up costs, so that main job and sub-jobs show values of work carried out both at that level and at all subordinate levels.

Budgets - The system attaches budgeted costs at every level, with a breakdown of year and period. A budget type identifies whether the values are associated with budget or forecast costs.

Budgets can be set at any job level. The application will automatically roll up budgets from a lower level through the job hierarchy. For greater flexibility, it is possible to record a rolled-up budget at a higher level - and then protect it from being overwritten during the automatic roll-up.

Budgets can also be produced against an expense type at each job level, with six budget levels created within any one budget code:

- Job level
- Sub-job level
- Task level
- Job / expenses level
- Sub-job / expenses level
- Task / expenses level

The system defaults to a user-defined budget code. However, further budgets or forecasts can be created for any one job.

Maintenance Data

Job Management does require some maintenance data to be set up, but in general this is a limited amount, and requires little attention after the initial set-up.

The Company Profile

Before any transactions are posted, key application controls must be set up. These include entering holding accounts and active interfaces, reserving expense types to particular transactions, and finalising batch control details.

Parameter Maintenance

Most of this data simply needs a code and a description to be entered. Data within this category includes:

Managers	Status codes
Locations	Budget codes
Regions	Transaction sources
Divisions	Transaction types
Industry codes	Variation reason codes
Profit / cost centres	Analysis codes
Charge codes	Expense types
Retention types	Uplift codes
Penalty clauses	Activity codes

Job Types

The job type dictates how the system processes the job, e.g. the valid expense codes for this job type, or any default text that is to appear on invoices.

Employee Details

Each employee on a job needs to be entered on the employee file. The required data includes employee number, name, etc., plus a grade.

Employee Grades

Each employee has an associated grade code, which can be associated with a price list to calculate costs and cost to complete.

Process Codes

Certain predefined processes, invoicing for example, can be set to change a job detail's status code automatically to a user-specified value when the process takes place.

Transaction Processing

The normal day-to-day processing in Job Management consists of:

- Building quotations
- Adding and amending jobs
- Processing transactions
- Producing invoices
- Releasing job costs and income from work in progress balance sheet accounts

Building Quotations

Quotations are built up using an identical hierarchical structure to jobs, with a header, a details screen, a hierarchy where jobs, sub-jobs and tasks can be entered, and transactions which can be entered at any level.

The similarity between a quotation structure and a job structure means that when you have built your quotation you can transfer it as a budget with the headers and details also being copied across. The transfer of details is flexible, and you may decide not to copy across at all.

No postings to GL are made. The estimated costs, which can be added for any level in the hierarchy, can build up using standard functions such as part lists and user-defined transactions.

Furthermore, price overriding and global mark-up changes are available for all types of transactions.

If you have the Sales Order Processing / Advanced Order Entry (SOP / AOE) link active, then, a parts list can also be attached to a Job Management quotation. The sales order that is created will be held as suspended, if the parts list is for a quotation.

The following steps show how the quotation feature within Job Management works:

Step 1

In building a quotation, you create a quotation header with quotation details.

Step 2

From the header you may add sub-job and tasks for the quotation. This is done on the quotation work with details screen.

Step 3

Behind each quotation detail (i.e. any header, sub-job or task number), you can add a number of expense types. These expense types are added and maintained on the quotation work with expenses screen.

Step 4

Behind each expense type, you can add a number of transactions. These transactions have a reference number and a description. These transactions are added and maintained on the quotation work with transactions screen. Global

mark-ups, selling prices and margins can be adjusted in order to arrive at a final quotation price.

There are a number of copy facilities available for quotations:

- Copy a closed quotation to a live job.
- Copy a sub-job / task to another sub-job / task within a quotation.
- Copy an existing quotation to a new quotation.
- Copy a job to a quotation.

Any quotation can be revised, and the original version and earlier revisions will be kept for reference. The system will always default to the latest revision, but other revisions can still be maintained. Revisions are automatically sequenced by date and time.

Adding and Amending Jobs

To create a new job it is necessary to define a job header and a main job at the job level with optional sub-jobs and tasks. A credit check on the customer may be carried out when the job is raised (determined by job type).

Necessary information on the job level includes that which is made available across a number of jobs for analyses such as project manager, division, region etc. Other information specific to the job, such as estimated start and finish dates are also entered in the job details at job level.

Those jobs designated as fixed price offers you an opportunity to enter stage payment details. If designated as non chargeable or time & material (T&M), the stage payment screen is not revealed.

Processing Transactions

Costs attributable to those jobs which are not covered by the automatic update facilities - timesheets, other application interfaces and invoicing - are catered for by direct entry. User-defined transaction types allow for flexible analysis of items such as adjustments and cash payments.

Timesheets

Although many organisations enter timesheet data on a weekly / monthly cycle, it is possible to record the entries on a daily basis, with a periodic update of the job costs. The data recorded is used to update actual labour costs on both chargeable and non-chargeable jobs, and calculate invoice values on T&M jobs. Timesheets can be reviewed and amended before posting.

Employee Expenses

Employee expenses may be entered either from within Job Management, or from Accounts Payable. If entered from Accounts Payable, the transactions are posted to Job Management when the invoice is accepted. The employee mileage and claim amount to be charged can be automatically calculated against a job, as required.

Standing Costs

Standing costs simplify the posting of costs against a job number, where the cost could be deemed 'regular' or 'standard'. For instance, rent of machinery could be a standard cost with a fixed currency, an expense type, a transaction type and charge code. Instead of entering these details in the direct transactions option, it

is possible to choose the standing reference and enter the amount. Each standing reference has an associated set of job-numbers (i.e. job / sub-job / task). Reversible standing costs can also be set up to deal with periodic accruals.

Releasing Job Costs and Income

Costs and income can be released from GL work in progress by varying user-defined methods including:

- Automatic release without user intervention.
- User-controlled release. The release facility allows a range of control from automatically releasing all values for a job to the defined release of part of one transaction.
- Additional control can be put in place so that the use of the user-controlled release outlined above can apply only to closed jobs.

Invoicing

Invoicing can be achieved via the three methods listed below. Some of the invoicing features include:

- Multiple currencies within a job.
- Multiple delivery addresses.
- Support of statement accounts, extended terms etc.
- The facility to review invoices, change invoice values and test before final printing.
- Reprint facilities.

Invoice Methods

Stage Payments

These are automatically generated from jobs where the job type is defined as being a fixed price job. An invoice is generated whenever the date of a stage payment is reached, or passed. Stage payments can depend on either of the following:

- When a specified date is reached.
- When specified activities are complete.

Time & Materials

These are generated from jobs where the job detail is defined as 'time and materials'. An invoice is generated whenever a rechargeable cost transaction is recorded against such a job.

Direct Entry

There are facilities that enable you to enter an invoice directly. These only apply to fixed price jobs. Sundry credit notes can also be issued for fixed price jobs.

Enquiries

There are several enquiry routes within Job Management.

Company Profile Enquiry

This displays all the company profile information.

Parameter Enquiries

There are enquiries for all the application reference data entered within the system. They consist of output-only versions of the parameter maintenance function.

Main Job Enquiry

The main job enquiry requires selection of job enquiry by customer or job number - subject to status code.

The job details enquiry provides standard information on the set up details as well as values for costs, budgets, variances and invoices to date. Similar information can be provided on both sub-job and task enquiry details. Further financial details are displayed for rolled-up and this level details. This information can be reviewed in more detail, i.e.:

- Commitments (Purchase Management or Inventory Management)
- Invoices
- Price variances
- Stage payments
- Cost breakdown this level
- Indented cost breakdown
- Alternative budgets

Session Enquiries

The front-end screen requests an entry via a transaction reference number or a session number. The enquiry by session number has drill-down facilities, to view transaction, source and destination details.

Other Enquiries

The Job Enquiry By Asset Number identifies all job details that are associated with a particular asset number, where Fixed Assets is active.

The Quotation Enquiry shows the breakdown of costs against a quotation with the associated mark-up percentages needed to arrive at a proposed selling price.

The Parts List Enquiry facilitates an enquiry on the status of inventory items associated with a job together with the stock availability set against individual items (where SOP is active).

If the Production interface is being used, an enquiry will reveal all production orders associated with a job. The status of the order can be viewed and details of input (component) availability displayed.

Reports

Generally, standard reports are printed on standard listing paper with 132 print positions and 6 lines to the inch.

Parameter Reports

You can produce a report of all the existing parameter entries. If, for example, you produce a report of all the job types for a company, the report will include all valid expense types and the flags set for that job type (e.g. invoice hours text etc.).

Analysis Reports

There are three principal analysis reports that can be created. The options are Job Analysis, Sub-job Analysis and Transactions by Task.

Each report can be generated for status of open jobs, closed jobs or both. These reports detail for a range of parameters. For instance, a report can be generated that involves all job numbers, or is specific to a particular task.

Each report displays details on actual revenue to date, any commitments outstanding (i.e. from Purchase Management and Inventory Management).

- Actual costs (total and current month)
- Actual budgets (YTD and total)
- Variance value and percentages

Transaction By Task enables all your transactions that have been entered to date to be reviewed. It displays the expense type, reference numbers and value, with internal text attached.

Returned Assets Report

This report identifies the return, via the Job Management / Customer Returns interface, of any serial-controlled Inventory item that has previously been capitalised via Job Management.

Transaction Processing Report

When you enter costs through Job Management, two reports are produced automatically; a posting validation error report and a GL interface file validation report.

When you create timesheet transactions, a timesheet reconciliation report is produced. This lists employees whose timesheets have not been entered, or whose hours were less than the standard weekly hours.

When costs are posted from through Job Management to the General Ledger posting validation report and a General Ledger File validation report are produced.

For timesheet transactions, a timesheet reconciliation report is produced. This produces a report of employees whose timesheets have not been entered or whose hours have been less than their normal weekly hours as defined in the employee details maintenance.

The interface file validation report for the General Ledger contains details of sessions for Job Management cost transactions. The following details are outlined in this report.

Session Details:

- Session number.
- Company.
- User and device.
- Date & time.

Header Details:

- Transaction type.
- Document reference.
- Posting period.
- Description of job detail.
- Company.
- Source account.
- System session number.
- Currency code of transaction.
- Prime control amount.

Line Details:

- The GL account.
- Prime value control (of line).
- Base value.
- Memo quantity.
- Secondary analysis codes.
- Source system for accounts.



Warning: Warnings are also highlighted for failed postings to GL accounts via the mask.

Invoice Reports

In addition to the final invoices and credit notes, appropriate interface file validation reports are automatically produced. A report for generated invoices needing review and a report of all generated invoices are also automatically created.

Fixed Assets Report

To validate the transfer of assets, three reports are produced automatically:

- Asset creation report - a report for every asset that is being created for the first time.
- Asset postings report - a report of the costs being posted to Fixed Assets.

- Exception report - a report of any errors that would occur within creation and/or amendment of Fixed Assets if all transactions were to be processed. It also flags any transaction causing such an error so that it is not processed in the actual transfer.

A summary report is generated for any released transactions posted to General Ledger and Fixed Assets.

Prerequisites

Two applications are prerequisites for the running of Job Management:

- General Ledger
- Accounts Receivable

Implementation Guide

1. Install GL (General Ledger) and Accounts Receivable. These are prerequisite to the installing of Job Management.
2. If you are intending to make use of the Job Management - General Ledger posting interface, then you must set up the GL chart of accounts with the accounts which will be receiving Job Management postings.
3. Other applications you may wish to interface with should be installed. These are not prerequisite to the running of Job Management, and can therefore be installed after you have begun using Job Management. They may include:
 - Fixed Assets
 - Inventory Management
 - Purchase Management
 - Accounts Payable
 - SOP (Sales Order Processing) / AOE (Advanced Order Entry)
 - Production
 - Service Management
 - Customer Returns
4. Install Job Management, setting up library lists in accordance with the application installation instructions.
5. Go into the Company Profile maintenance option from the Utilities menu to set up your company. The system prevents you seeing the third details screen, because this requires the entry of parameters you have not yet had the opportunity to set up!
6. Your company now exists, and you must set up the parameters you intend to use for processing and analysis. These are found on the Parameter Maintenance menu.

If you do not want to use all of the parameter details for a job detail, you can set up your parameters so that a blank code is valid.
7. Re-enter the Company Profile maintenance option and enter the details screen which takes the parameter and holding account details. You can now enter your default parameters.

Do not change the batch number parameters within the company profile once you have started using Job Management.
8. If you are running the system on an AS/400, you must ensure that the Job Management subsystem is started before making transactions.

If the subsystem is not started, transactions will not be lost but will not be processed until the subsystem is started.
9. You can now start a job or quotation.

System Generated Data

When you set up a company profile, the system automatically creates some essential data within Job Management. The data created is as follows:

Status Codes

01 - New

99 - Closed



You will need to set up further status codes used by the system in conjunction with the process codes listed below.

Transaction Sources

IM - Inventory movement

JM - Job Management

PL - Accounts Payable

RC - Release cost

RI - Release income

RS - Release standing cost

SC - Standing cost

SR - Standing cost reversal

Process Codes

IVD - Invoiced

PDS - Partial despatch

RCV - Goods received

RTN - Return generated

SOC - Sales order created

TDS - Total despatch

Fixed Asset / General Ledger Account Types

090 - Job Management postings

Production Reschedule Policies

These codes protect the four order statuses used by Job Management. Materials Requirements Planning will be prevented from changing Job Management orders. The codes are set up in the Maintain Parameters option in the Production application, under parameter type 'WTYP'.

Planned: 41J

Confirmed: 42J

Released: 43J

Active: 44J



Note: Production orders generated via Job Management have a planning filter of J.

AOE Quotation Type

Q - Job Management quotation.

This code is set up via the Descriptions option in the Inventory Management module, under parameter type QTNO.